SEMTM0051 Foundations of AI
SAMPLE PAPER

This is intended as an example of what an exam paper will look like, we have tried to make it similar in scope and difficulty, every time we set a sample paper students complain afterwards that the sample paper was misleading because it was easier than the real paper, but really, we've tried to make them the same level!

This paper contains two parts.

The first section contains 25 short questions.
Each question is worth two marks and all should be attempted.

The second section contains three long questions.
Each long question is worth 10 marks.
The best two long question answers will be used for assessment.

The maximum for this paper is 70 marks.

A sheet of formulae is provided at the end of the paper.

Other Instructions:
Only non-programmable calculators may be used.
Calculators must have the Faculty of Engineering Seal of Approval.

TURN OVER ONLY WHEN TOLD TO START WRITING

Section A: short questions – answer all questions

Q1. A regression model is trained by least squares to predict y with a single constant c, that is, the model has no input. The training targets are $y = 2, 4, 6, 8, 10$. What value of c minimises the squared-error loss?

Q2. In gradient descent the parameters are updated as $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L$. What goes wrong if the learning rate η is set too small, and what goes wrong if it is set too large?

Q3. Explain how the model $y = a + b \sin x + c \cos x$ can be fitted using linear regression, even though it is not linear in x.

Q4. What is 'softmax'?

Q5. A three-class classifier outputs logits $(1, 1, 0)$. Compute the softmax probabilities to three significant figures; the softmax formula is on the formula sheet.

Q6. Why is the output of a binary classifier passed through a sigmoid function before computing the loss?

Q7. A binary classifier assigns probability 0.9 to the positive class for a particular input. Compute the cross-entropy loss for this input if the true label is positive, and if the true label is negative; the formula is on the formula sheet.

Q8. What do the extra layers of a deep neural network add that makes it better for complicated data?

Q9. Sketch a sigmoid, $\sigma(x)$.

Q10. An affine layer takes an input row $\mathbf{x}$ with 50 coordinates and produces 10 outputs according to $\mathbf{a} = \mathbf{x}\mathbf{W} + \mathbf{b}$. How many trainable parameters does it have, including the biases?

Q11. A network is built from two linear layers with no nonlinearity in between: $\mathbf{y} = \mathbf{x}\mathbf{W}_1\mathbf{W}_2$. Explain why this is no more expressive than a single linear layer. For simplicity biases are zero in this question.

Q12. A neuron has preactivation $a(x) = 2x - 1$ and uses ReLU. Write the corresponding activation as a piecewise-defined function and sketch it.

Q13. What is an epoch? A training set has $10\,000$ examples and the minibatch size is 100; how many parameter updates are made in one epoch?

Q14. What is underfitting? Describe what you would see in the training and validation losses if a model is underfitting.

Q15. What is an autoencoder? Why do we use them?

Q16. How does a variational autoencoder differ from an autoencoder?

Q17. Describe how a new data point is generated from a trained variational autoencoder.

Q18. A sparse autoencoder is trained with the loss $\|\mathbf{x} - \mathbf{W}_d\mathbf{h}\|^2 + \lambda\|\mathbf{h}\|_1$ where $\mathbf{h} = \mathrm{ReLU}(\mathbf{W}_e\mathbf{x} + \mathbf{b})$ and the hidden dimension is much larger than the input dimension. What does the $\lambda\|\mathbf{h}\|_1$ term encourage, and why is the hidden layer chosen to be so wide?

Q19. A 64×64 image is passed through a convolution with a 3×3 kernel, stride 1 and padding of one pixel on each side, followed by 2×2 max pooling. What are the spatial dimensions after each operation?

Q20. How many trainable parameters does a convolutional layer with eight input channels, 32 output channels and a 5×5 kernel have, including biases?

Q21. A convolutional layer has far fewer parameters than a fully connected layer acting on the same image. Explain why.

Q22. In attention, each token produces a *query*, a *key* and a *value*. Using the analogy of looking something up in a dictionary or database, explain the role of each of the three.

Q23. What is the difference between self-attention and cross-attention?

Q24. What is a causal mask in self-attention, and why is it needed when training a model to predict the next token?

Q25. What is multi-head attention, and why are several heads used rather than one?

Section B: long questions – answer two questions

Q1. This question is about backpropagation. Consider the network

$$\mathbf{a}_1 = \mathbf{x}\mathbf{W}_1, \qquad \boldsymbol{\phi}_1 = \mathrm{ReLU}(\mathbf{a}_1), \qquad f = \boldsymbol{\phi}_1\mathbf{W}_2, \qquad L = (f-y)^2,$$

where f is short for $f_\theta(\mathbf{x})$ and with

$$\mathbf{x} = \begin{pmatrix} 2 & 1 \end{pmatrix}, \qquad \mathbf{W}_1 = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \mathbf{W}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad y = 1,$$

and all biases are fixed at zero.

(a) Compute the forward pass: $\mathbf{a}_1$, $\boldsymbol{\phi}_1$, f and L. **[3 marks]**

(b) Compute $\partial L/\partial f$ and hence $\partial L/\partial \mathbf{W}_2$. Note that $\partial L/\partial f = 2(f-y)$, and that f depends on the entry $W_{2,i}$ through the term $\phi_{1,i}W_{2,i}$. **[3 marks]**

(c) Compute $\partial L/\partial \boldsymbol{\phi}_1$ and then $\partial L/\partial \mathbf{a}_1$. Note that $\mathrm{ReLU}'(a) = 1$ if $a > 0$ and 0 when $a < 0$. **[2 marks]**

(d) Without doing any further calculation, state which column of $\mathbf{W}_1$ will not change in the gradient descent update, and explain why in one or two sentences. **[2 marks]**

Q2. This question is about self-attention in transformers. A sequence of three tokens has embeddings

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \mathbf{x}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \mathbf{x}_3 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

A single-head causal self-attention layer uses query, key and value matrices all equal to the identity, $\mathbf{W}_Q = \mathbf{W}_K = \mathbf{W}_V = \mathbf{I}$, so that $\mathbf{q}_i = \mathbf{k}_i = \mathbf{v}_i = \mathbf{x}_i$. Ignore the $1/\sqrt{d}$ scaling.

(a) The raw score for token i attending to token j is the dot product $\mathbf{q}_i^\top \mathbf{k}_j$. Write down the 3×3 matrix of raw scores. **[3 marks]**

(b) Which entries of this matrix does the causal mask remove, and why? **[2 marks]**

(c) Compute the output of the attention layer at positions 1 and 2, to two decimal places. Position 1 can attend only to itself; position 2 has a two-way softmax over the scores S_{21} and S_{22}, and the output is the weighted sum of the values. **[3 marks]**

(d) For a sequence of n tokens, how does the cost of computing the attention scores scale with n? Explain briefly why this is a problem for long sequences. **[2 marks]**

Q3. This question is about probabilistic and sparse autoencoders. A variational autoencoder with encoder $q_\phi(\mathbf{z} \mid \mathbf{x})$, decoder $p_\theta(\mathbf{x} \mid \mathbf{z})$ and prior $p(\mathbf{z})$ is trained by maximising

$$\mathcal{L}(\theta, \phi) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\big[\ln p_\theta(\mathbf{x} \mid \mathbf{z})\big] - \mathrm{KL}\big(q_\phi(\mathbf{z} \mid \mathbf{x}) \,\|\, p(\mathbf{z})\big).$$

(a) Explain in words what each of the two terms rewards or penalises, and what would go wrong if the second term were left out. **[3 marks]**

(b) For a one-dimensional latent with $q(z \mid \mathbf{x}) = \mathcal{N}(\mu, \sigma^2)$ and $p(z) = \mathcal{N}(0, 1)$, the KL term is $\frac{1}{2}(\mu^2 + \sigma^2 - 1 - \ln \sigma^2)$. Compute it for $\mu = 1$, $\sigma = 2$, to two decimal places. What values of μ and σ make it zero? **[2 marks]**

(c) Evaluating the first term requires drawing a sample $z \sim \mathcal{N}(\mu, \sigma^2)$. Explain why backpropagation cannot pass through a random sampling step to reach μ and σ, and how writing $z = \mu + \sigma\epsilon$ with $\epsilon \sim \mathcal{N}(0, 1)$, the reparameterisation trick, solves the problem. **[3 marks]**

(d) A sparse autoencoder is trained on the hidden activations of a language model. Explain what a “feature” of the sparse autoencoder is, and describe one way you could check whether a given feature is interpretable. [**2 marks**]

Formulae

Linear regression. Optimal parameters: $\boldsymbol{\theta}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$, provided $\mathbf{X}^\top \mathbf{X}$ is invertible. Here $\boldsymbol{\theta}$ is the complete parameter vector, including the intercept, and when an intercept is fitted the design matrix $\mathbf{X}$ includes a column of ones.

Matrix inverse. $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \dfrac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$, assuming $ad - bc \neq 0$.

Sigmoid and softmax. $\sigma(x) = \dfrac{1}{1 + e^{-x}}, \qquad \operatorname{softmax}(\mathbf{a})_i = \dfrac{e^{a_i}}{\sum_j e^{a_j}}.$

Cross-entropy loss. Binary, with p the predicted probability of the positive class and $y \in \{0, 1\}$: $L = -y \ln p - (1 - y) \ln(1 - p)$. Multi-class with true class c: $L = -\ln \operatorname{softmax}(\mathbf{a})_c$.

Derivatives. $\sigma'(x) = \sigma(x)(1 - \sigma(x))$. $\mathrm{ReLU}'(a) = 1$ if $a > 0$ and zero when $a < 0$; often software will define the derivative as zero for $a = 0$ as well, but this is just a convention and not mathematically coherent.

2D convolution. $a_{x,y} = \displaystyle\sum_{\delta_x=-1}^{1} \sum_{\delta_y=-1}^{1} h_{x+\delta_x,\, y+\delta_y}\, W_{\delta_x, \delta_y}.$

Momentum. $\mathbf{v} \leftarrow \beta \mathbf{v} + \mathbf{g}, \quad \mathbf{w} \leftarrow \mathbf{w} - \eta \mathbf{v}.$

Adam. $\mathbf{m} \leftarrow \beta_1 \mathbf{m} + (1 - \beta_1)\mathbf{g}, \quad \mathbf{s} \leftarrow \beta_2 \mathbf{s} + (1 - \beta_2)\mathbf{g}^2, \quad \mathbf{w} \leftarrow \mathbf{w} - \eta \dfrac{\mathbf{m}}{\sqrt{\mathbf{s}} + \epsilon}$ (bias corrections omitted).

Scaled dot-product attention. $\operatorname{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \operatorname{softmax}\!\left(\dfrac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right)\mathbf{V}$, with softmax applied row-wise.

KL divergence between Gaussians. $\mathrm{KL}\big(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0, 1)\big) = \tfrac{1}{2}\left(\mu^2 + \sigma^2 - 1 - \ln \sigma^2\right).$

Evidence lower bound. $\ln p_\theta(\mathbf{x}) \;\geq\; \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\big[\ln p_\theta(\mathbf{x} \mid \mathbf{z})\big] - \mathrm{KL}\big(q_\phi(\mathbf{z} \mid \mathbf{x}) \,\|\, p(\mathbf{z})\big).$

END OF PAPER